1 Defining the LUTs

For reducing the computational burden, functions such as $\sqrt{x}$ and $\mathrm{tg}^{-1}(x)$ were implemented through the use of look-up tables (LUTs). Nonetheless, tabling these functions for a large variety of x values tends to compromise immense portions of the FPGA storage space, and thus it is prohibitive. To overcome this issue, the mentioned functions were split into line segments given by:

$$\tilde{f}(x) = mx + n, \text{ for } x_a \leq x < x_b \tag{1}$$

where $[x_a, x_b]$ is the interval for which $\tilde{f}(x)$ is valid. Thus, instead of storing the different values of $f(x)$, we can table the values of m and n for the different segments of the function. This approach requires some extra computations, say it $\tilde{f}(x) = mx + n$, but significantly reduces the required storage space.

In this case, coefficients for the line segment in the interval $[x_a, x_b]$ is given by:

$$m = \frac{f(x_b) - f(x_a)}{x_b - x_a} \tag{2}$$

$$n = f(x_a) - mx_a \tag{3}$$

The precision of the representation will depend on the amount of segments used along the domain, something that is highlighted in Fig. 1 for the approximation of the square-root (SQRT) function. When we consider four line segments in the interval [1, 250], for instance, the error is lower than 5% only when the argument x is greater than 56.9. When we increase the number of segments, the region for which the error is always bellow the 5% widens, at the cost of larger storage requirement. Of course, for such a narrow range as [1, 250], the number of segments seams not to be a big deal, but when it comes to the real range considered for the system variables, i.e. [0, 32768], the number of segments to obtain small errors such as 5% throughout the domain might be overwhelming.

For guaranteeing small error while using the smallest number possible of segments, the digital implementation considers segments with non-uniform intervals, such it is presented in Fig. 2.

2 Number of intervals for the SQRT

For determining the number of interval, it was defined an limit error of 1%.

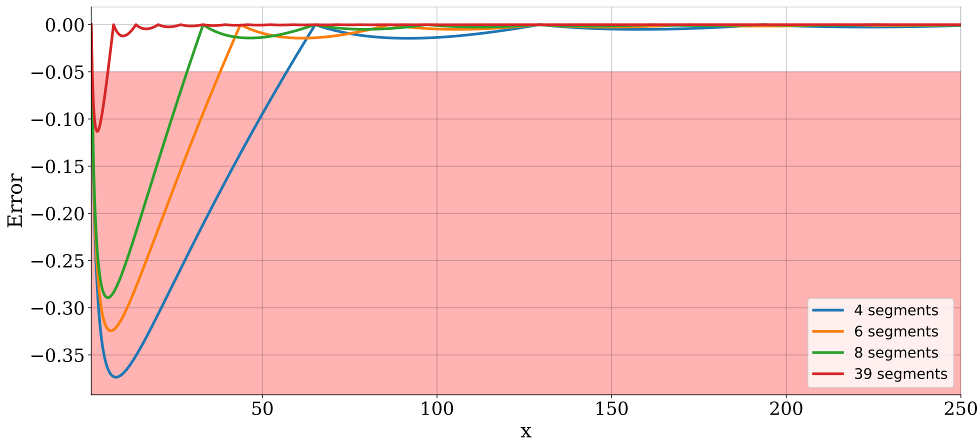

Figure 1: SQRT approximation error in the interval [1, 250] for different number of segments.

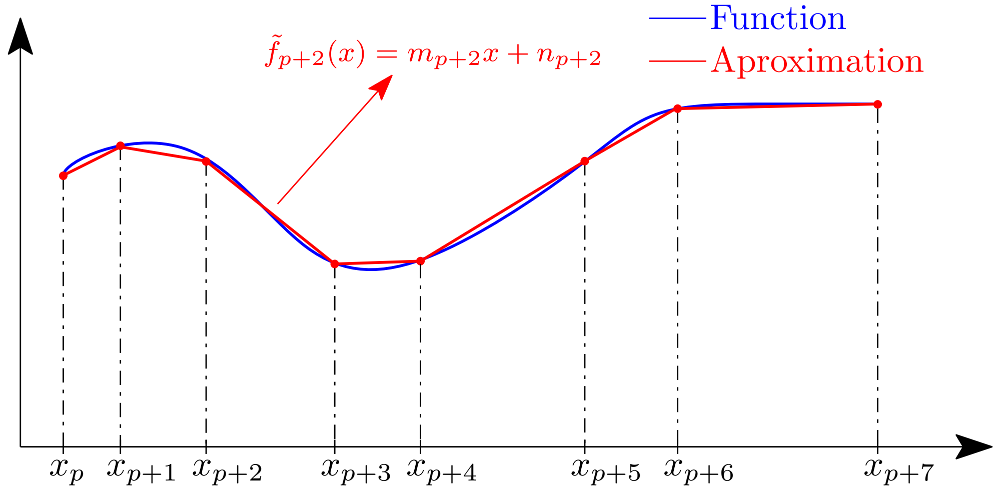

Figure 2: Approximation methodology.

Interval	number of intervals
$[0.1, 1]$	13
$[1, 10]$	13
$[10, 100]$	13
$[100, 1000]$	13
$[1000, 10000]$	13
$[10000, 32000]$	4
Total	69

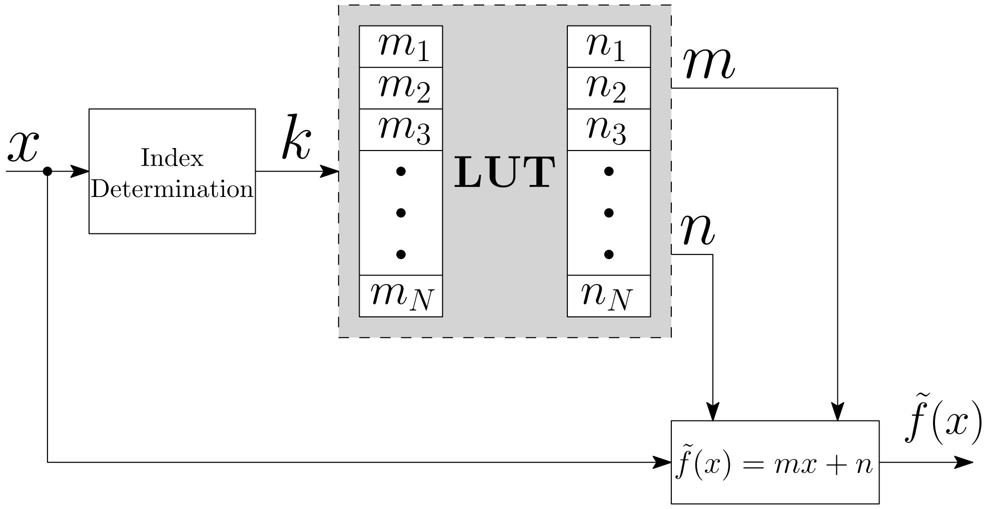

Figure 3: Look-up table implementation.

3 Number of intervals for the ATAN

For determining the number of interval, it was defined an limit error of 1%.

Interval	number of intervals
$[0, 1]$	7
$[1, 2]$	4
$[2, 10]$	10
$[10, 100]$	7
$[100, 1000]$	7
$[1000, 32000]$	2
Total	37